

AI EPC Platform - Engineering Specification

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Project : Alpha Refinery

Vendor : Siemens

Equipment : Compressor

Issue Date : 2025-04-20

Project Overview

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The project includes utility power integration, UPS systems, cooling systems, and fire protection.

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The facility shall comply with Uptime Institute Tier III recommendations.

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Cooling System

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Each cooling unit shall include vibration monitoring sensors.

Chilled water pumps shall operate with redundant motors.

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Cooling redundancy shall be verified during commissioning.

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The cooling system shall provide N+1 redundancy.

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AI EPC Platform - Engineering Specification

CRAH units shall support automatic failover.

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Cooling redundancy shall be verified during commissioning.

Cooling equipment shall support SNMP monitoring.

Chilled water pumps shall operate with redundant motors.

CRAH units shall support automatic failover.

Supply air temperature shall remain between 18°C and 27°C.

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Electrical System

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Electrical panels shall comply with IEC standards.

UPS capacity shall be 500 KVA.

Power Distribution Units shall provide branch circuit monitoring.

AI EPC Platform - Engineering Specification

Dual power feeds shall be provided for critical equipment.

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Generator backup shall support 72 hours of continuous operation.

Emergency shutdown shall disconnect non-critical loads.

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AI EPC Platform - Engineering Specification

Fire Protection

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VESDA detection shall be installed.

Emergency evacuation procedures shall be displayed.

Fire alarm panels shall integrate with the BMS.

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Fire alarm panels shall integrate with the BMS.

Emergency evacuation procedures shall be displayed.

FM200 suppression shall protect all server rooms.

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Fire alarm panels shall integrate with the BMS.

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Smoke detectors shall be installed in all electrical rooms.

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Network Infrastructure

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Network monitoring shall integrate with the NOC.

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All network racks shall use Cat6A cabling.

Core switches shall support redundant uplinks.

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Commissioning

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Integrated System Testing shall be completed before handover.

Generator synchronization tests shall be completed.

Integrated System Testing shall be completed before handover.

Commissioning reports shall include all test results.

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Commissioning reports shall include all test results.

Cooling performance shall be verified under full load.

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Generator synchronization tests shall be completed.

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Quality Assurance

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Vendor documentation shall be reviewed before installation.

All materials shall comply with project specifications.

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All construction activities shall follow QA/QC procedures.

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AI EPC Platform - Engineering Specification

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